Md Shoaib Mahmud

Email: shoaib09mahmud@gmail.com • Phone: +880 1734 355994 • LinkedIn • Google Scholar

Research Interests

My research develops optimization and machine learning methods for resilient networked systems.

Specific interests: supply chain and infrastructure resilience; graph neural networks for transportation networks; robust and stochastic optimization under uncertainty; prescriptive analytics for disaster response.

Education

B.Sc. in Industrial and Production Engineering Feb 2020 – Apr 2024

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

CGPA: 3.77 / 4.00 **Rank:** 4 of 51

Relevant Coursework: Operations Research, Probability and Statistics, Modeling and Simulation, Supply Chain Management.

Undergraduate Thesis: *Reverse Logistics Network Design for the Semiconductor Supply Chain*

- Developed a mixed-integer linear programming model for semiconductor reverse logistics, coordinating forward and reverse flows under capacity and routing constraints using IBM ILOG CPLEX.
- Designed a quick-response recovery network for testing and redeploying functional chips to reduce new-facility reliance and shorten lead times.

Publications

Journal Articles

[J1] **Mahmud, M. S.**, Ikbal, M. N., Sen Gupta, H., Sanderson, D., & Biswas, S. (2026). Graph neural networks for seismic vulnerability assessment and retrofit prioritization of transportation networks. *Reliability Engineering & System Safety*, Article 113006. *DOI: 10.1016/j.ress.2026.113006*

[J2] Sen Gupta, H., **Mahmud, M. S.**, & Gupta, A. (2026). From calibrated risk to actionable response: A prescriptive analytics pipeline for urban flash floods. *Resilient Cities and Structures*, 5(2), 111–125. *DOI: 10.1016/j.rcns.2026.03.005*

[J3] Zaman, T. A. B., **Mahmud, M. S.**, Sen Gupta, H., Harati, M., & van de Lindt, J. W. (2026). Quantifying the cost of delay in floodplain property buyouts. *Sustainability*, 18(6), 2675. *DOI: 10.3390/su18062675*

Manuscripts Under Review

[U1] **Mahmud, M. S.**, Gupta, A., Anindya, Z. M. T., Maitry, A., & Sen Gupta, H. Robust allocation of risk-reduction resources under vulnerability uncertainty. Under review at *Reliability Engineering & System Safety* (manuscript ID: JRESS-D-26-02793).

[U2] Sen Gupta, H., Saad, S. I., **Mahmud, M. S.**, & Biswas, S. (2025). Equity-aware ambulance dispatch after earthquakes. Under review at *Safety Science* (manuscript ID: SAFETY-D-25-03429).

[U3] Alam, S. T., Ahmed, T., Haque, T., **Mahmud, M. S.**, & Kaisar, T. I. Profit-oriented pricing combinations using RSM-ANN. Under review at *Discover Applied Sciences*.

[U4] Hossain, N. U. I., Ikbal, M. N., **Mahmud, M. S.**, Sen Gupta, H., Duarte, J. A., & Jaradat, R. (2026). Equity-aware interpretable policy rules for personalized hurricane risk communication. Under review at *International Journal of Disaster Risk Reduction* (manuscript ID: IJDRR-D-26-00493).

Conference Proceedings

[C1] **Mahmud, M. S.**, Saad, S. I., & Miazi, M. H. (2025). Data-driven application of the newsvendor model for retail inventory optimization. *Proc. 8th IEOM Bangladesh International Conference*, Dhaka.

[C2] Saad, S. I., **Mahmud, M. S.**, Hossain, M., & Rahman, A. Reducing delivery time in electric vehicle routing problems through adaptive partial recharging. *Proc. 4th International Conference on Innovations in Data Analytics*, Kolkata.

[C3] Debnath, A., & **Mahmud, M. S.**(2025). Dynamic staffing optimization for institutional dining services: An Erlang C queueing approach. *Proc. 8th IEOM Bangladesh International Conference*, Dhaka.

[C4] Rubaet, M., **Mahmud, M. S.**, Mukit, M. U. A., Haque, M. S., Mithu, M.-A.-I., & Sakib, M. I. A. (2025). The systematic design and manufacturing of "Mr. Money Manager". *Proc. 8th IEOM Bangladesh International Conference*, Dhaka.

[C5] Alam, S. T., Sagor, E., Ahmed, T., Haque, T., **Mahmud, M. S.**, Ibrahim, S., & Shahjahan, O. (2023). A comparative analysis of the assignment problem. In *Big Data Innovation for Sustainable Cognitive Computing* (pp. 125–142). Springer Nature.

[C6] Alam, S. T., Ahmed, T., Haque, T., & **Mahmud, M. S.**(2022). A comparative analysis of balanced transportation problems. *Proc. ICMEAS 2022*, MIST, Dhaka.

Research Experience

Research Affiliate, RISE Lab, Colorado State University Pueblo Jan 2025 – Present

Supervised by Dr. Himadri Sen Gupta

- Developed **DualGINE-HGT**, a physics-supervised dual-branch graph neural network for seismic vulnerability assessment on the Seaside, Oregon road network; achieved MAE 0.0158 and R^2 0.9777, improving MAE by 20.5% over XGBoost baselines.
- Developed a two-stage robust optimization model with ML-informed link vulnerability for post-tornado power-grid restoration.
- Co-developed prescriptive-analytics models for disaster response, including a calibrated risk-to-response pipeline for 6,137 Texas flash-flood incidents using regularized logistic regression, isotonic calibration, and Gurobi portfolio optimization, and an equity-aware time-expanded linear program for post-earthquake ambulance dispatch in Shelby County, Tennessee that improved minimum zone-level service by 9.17 percentage points and reduced the population-weighted Gini coefficient by approximately 37%.

Teaching Experience

Lecturer November 2024 – Present

Department of Industrial and Production Engineering, Military Institute of Science and Technology (MIST)

- **Theory courses:** IPE 101 – Introduction to Industrial and Production Engineering, IPE 107 – Engineering Economy, MATH 215 – Numerical Analysis, IPE 317 – Ergonomics and Safety Management, IPE 419 – Modeling and Simulation.

- **Sessional:** IPE 200 – Engineering Graphics and CAD Sessional, IPE 206 – Probability and Statistics Sessional, IPE 306 – Operations Research Sessional, IPE 318 – Ergonomics and Safety Management Sessional.
- Mentor and advise students on class projects, undergraduate theses, and career planning, with emphasis on applying industrial engineering tools to real-world problems.
- Contribute to curriculum design by incorporating modern optimization software and simulation practices, aligning coursework with current industry standards and research frontiers.

Presentations and Guest Lectures

- "Graph Neural Networks for Seismic Vulnerability Assessment and Retrofit Prioritization of Transportation Networks." Guest lecture, EN 591: Predictive Analytics, RISE Lab, Colorado State University Pueblo, 30 April 2026.

Technical Skills

Machine Learning	PyTorch, PyTorch Geometric, graph neural networks (GINE, Heterogeneous Graph Transformer), probability calibration (isotonic, temperature scaling), scikit-learn, XGBoost, LightGBM, ensembling and spatial cross-validation.
Optimization	Gurobi, IBM ILOG CPLEX; mixed-integer linear programming; two-stage and robust optimization; stochastic programming; network and resource-allocation models.
Programming, Data	Python (NumPy, Pandas, SciPy, NetworkX), C, C++.
Simulation, Stats	SimPy, Arena, Minitab, SPSS, Power BI.
Writing	LaTeX, Microsoft Office, Mendeley, Zotero.

Honors and Awards

Dean's List of Honor, MIST 2020–2024

- Awarded for sustained CGPA above 3.75.

University Merit Scholarship, MIST 2021–2024

- Awarded for top-three class positions across consecutive semesters.

Top-Performing Trainee Feb 2023

- Recognized during the British American Tobacco Bangladesh Industrial Attachment Program.

Academic Service and Professional Experience

Editor, Technical Paper Volume 3, Department of IPE, MIST Jun – Oct 2025

- Led peer review and editorial coordination for the departmental research volume.

Director of Program, IEOM MIST Student Chapter Apr 2023 – Apr 2024

- Organized technical workshops on supply chain management and MATLAB.

Industrial Trainee, British American Tobacco Bangladesh Feb 2023

- Built Power BI dashboards over operational datasets.